

Indian Institute of Technology Kanpur

Department of Chemical Engineering

Mid-Semester Report

CHE251: Chemical Process Calculations Semester 2025-26-I

Group 2

Project Title:

Post-combustion CO₂ Capture from Coal Combustion Flue Gas
using Solvent Absorption and Stripper-Regenerator System

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1. Introduction

Climate change and rising greenhouse gas emissions have made CO₂ capture a critical area of research in chemical engineering. Among various industrial sources, coal-fired power plants are major contributors of CO₂ emissions. In India, coal remains the dominant fuel for electricity generation, which makes post-combustion capture technologies particularly relevant.

Post-combustion capture using chemical absorption is an established approach. Monoethanolamine (MEA) is widely studied because of its fast reaction with CO₂; however, considerations such as solvent degradation and regeneration energy are important in the overall process design. This report presents the basis, assumptions, flow schematics, and reaction mechanisms that form the foundation for subsequent mass and energy balances.

2. Basis of Calculations

All calculations are performed on a basis of **10,000 kg/h** of dry coal feed to the combustor. This basis represents a moderately sized industrial combustor and simplifies comparisons. The analysis assumes steady-state operation — a standard first approximation for process evaluation that keeps mass and energy flows constant with time.

3. Known Values and Assumptions

The assumptions used in subsequent calculations are listed with short rationale:

- **Ultimate analysis of coal:** C = 48.10%, H = 3.10%, N = 1.40%, S = 0.60%, O = 8.56%, Ash = 38.24% (wt%).
- **Complete combustion:** All C $\rightarrow$ CO₂, S $\rightarrow$ SO₂, and H $\rightarrow$ H₂O, giving an upper bound for flue gas production.
- **Air composition:** 21% O₂, 79% N₂ by volume.
- **Excess air:** 15% to avoid incomplete combustion.
- **Capture target:** 90% CO₂ capture efficiency.
- **Solvent:** 30 wt% aqueous MEA with lean loading = 0.2 mol CO₂/mol MEA and rich loading = 0.5 mol CO₂/mol MEA.
- **Hydrodynamics:** superficial gas velocity in absorber = 2 m/s.
- **Energy:** reboiler duty = 3.5 MJ/kg CO₂; latent heat of steam at 1 bar = 2257 kJ/kg.
- **Other:** negligible pressure drops and heat losses; FGD efficiency = 95% SO₂ removal.

4. Flow Diagrams

4.1 Overall Process

The overall schematic of the process is shown below. It highlights the sequence of major units including combustion, flue gas desulfurization (FGD), absorption, and solvent regeneration.

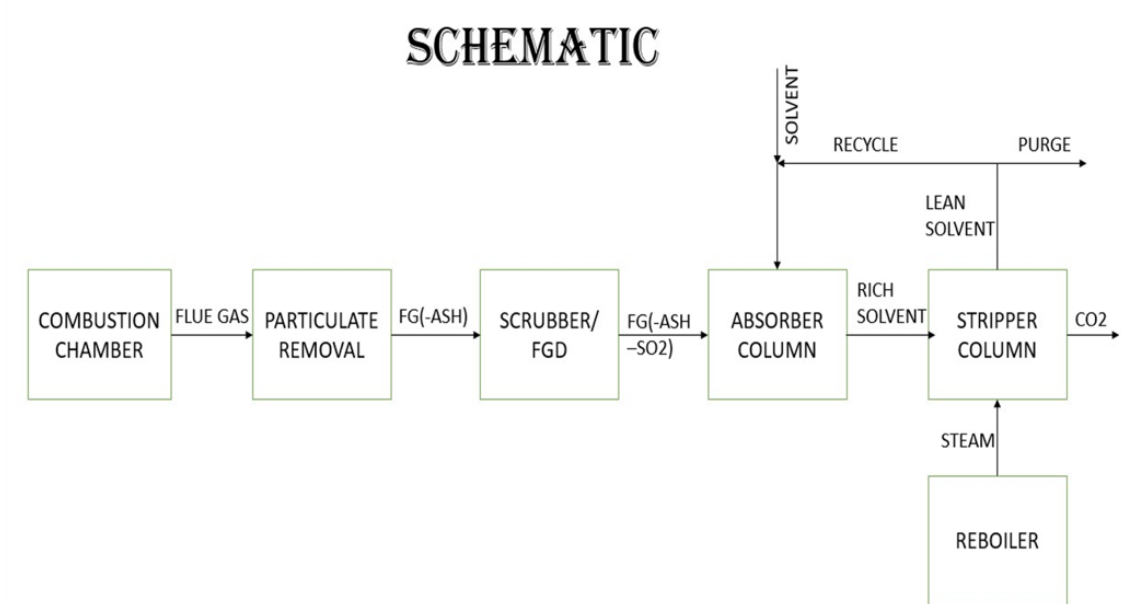


Figure 4.1: Overall process flow diagram

4.2 Combustion Chamber

The combustion chamber is the first stage of the process and is responsible for fuel oxidation producing flue gas composed primarily of N_2 , O_2 , CO_2 , H_2O and trace SO_2 , and carries fly ash. Coal reacts with air under controlled excess oxygen to generate flue gas containing mainly CO_2 , H_2O , N_2 , and trace SO_2 .

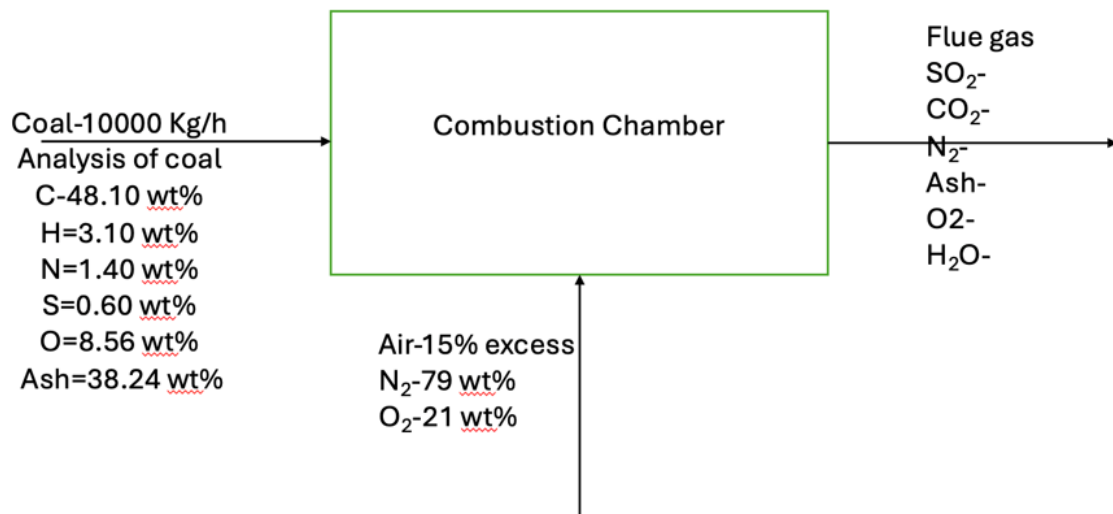


Figure 4.2: Combustion chamber schematic

4.3 FGD and Particulate Removal

Before entering the absorber, the flue gas passes through a particulate removal unit and an FGD system. Flue gas desulfurization using a limestone scrubber removes most SO₂ and particulate devices (ESP / baghouse) reduce ash to protect the solvent from contamination. This step reduces dust and captures most of the SO₂, preventing solvent degradation.

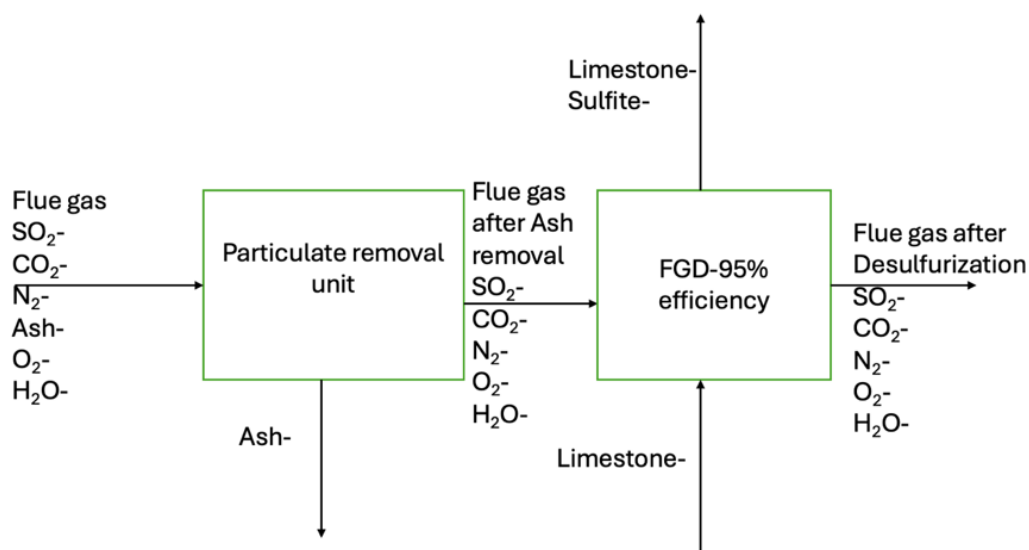


Figure 4.3: Flue gas desulfurization and particulate removal system

4.4 Absorber and Stripper

The absorber is a packed column contacting flue gas with lean MEA. CO_2 chemically reacts with MEA forming carbamate and ammonium species (rich solvent). The stripper applies heat (reboiler) to regenerate lean solvent and release concentrated CO_2 for compression. The absorber brings the treated flue gas into contact with lean MEA solvent, absorbing CO_2 and producing a rich solvent stream. The stripper, supplied with heat from the reboiler, regenerates the solvent and releases concentrated CO_2 for compression and storage.

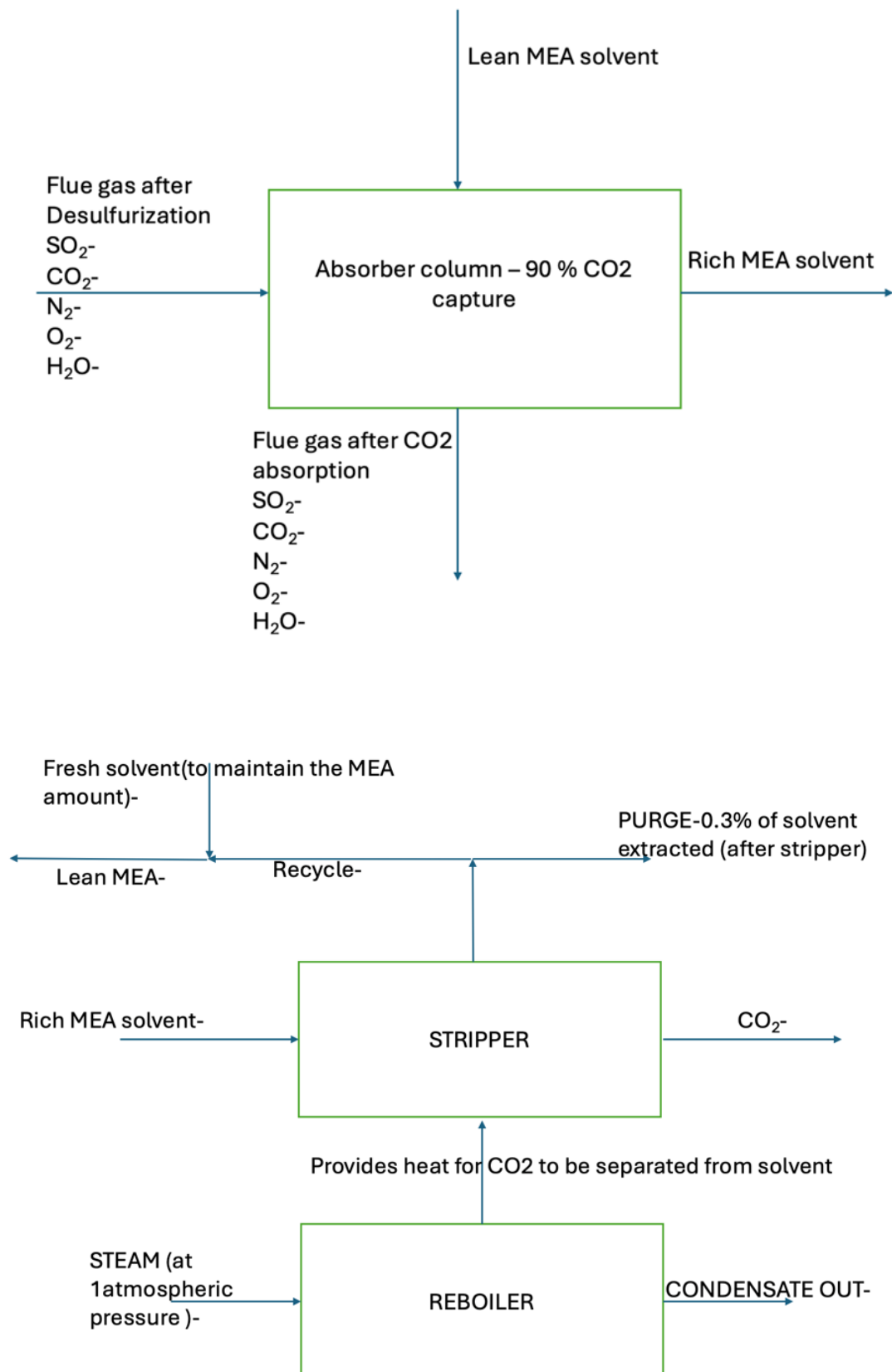


Figure 4.4: Absorber and stripper system

5. Reactions Involved

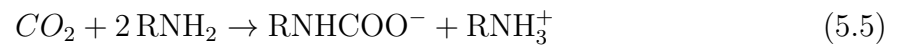
5.1 Combustion



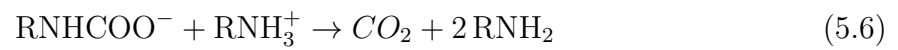
5.2 Flue Gas Desulfurization



5.3 CO₂ Absorption in MEA



5.4 CO₂ Desorption in Stripper



6. Conclusion and Future Work

This mid-semester report establishes the basis, assumptions, schematics, and reactions needed for a detailed simulation of CO₂ capture from coal flue gas. Next steps include component-wise material balances, column sizing, detailed energy calculations for the reboiler, and sensitivity analysis.